OCHMO Office of Research Assurance

Human Subjects Research Standard Operating Procedures(SOP)

SOP - 405: Human Subjects Research Determination

Number	Effective Date	Version Number	Previous Versions
405	8/14/2023	v.7.1	v.7

Purpose

HRP-405 defines the NASA IRB policy and processes related to determining whether certain types of projects, such as human in the loop (HITL) studies, quality improvements activities, engineering studies, or use of data/specimens are subject to oversight by the NASA Institutional Review Board (IRB).

NASA is an official signatory of the Common Rule, codified at 14 CFR part 1230 and applies the Food and Drug Administration (FDA) regulations 21 CFR parts 50, 56, 312, 812 (as applicable) when projects involve activities regulated by the FDA.

Prospective review and approval by an IRB is required for all projects that meet the regulatory definitions for *research* with *human subjects* (Common Rule) or a *clinical investigation* (FDA). Activities that do not meet the regulatory definitions are considered *"Not Human Subjects Research"* (NHSR) for which IRB oversight is not required. Making the determination of whether a project is or is not Human Subjects Research (HSR) is nuanced and context specific. Principal Investigators (PI) are encouraged to consult with NASA IRB staff whenever there is uncertainty.

Policy

1. Official NHSR determination Authority.
    a. Only the NASA IRB Staff, the NASA IRB Chair, or the convened NASA IRB are authorized to make an ***official IRB determination*** that an activity does or does not involve human subjects research (i.e., Not Human Subjects Research (NHSR)). While the federal regulations do not mandate that researchers obtain official NHSR documentation from the IRB, other NASA review boards (e.g., TRR, TSS, Safety) may require formal documentation.
        1. NASA IRB's determination that an activity is human subjects research supersedes the determination of another institution or agency's IRB when NASA is *engaged.*
        2. NASA IRB may make an NHSR determination based on information provided in a full-board/expedited/exempt submission.

3. The NASA IRB does not have authority to grant retroactive approval for a project meeting the definitions of *research* with *human subjects*.
4. When needed, the official NHSR determination should be sought prior to initiation. If a project is determined to constitute Human Subjects Research (HSR), prospective review and approval by the IRB must be sought before research activities begin.
5. NASA IRB has the authority to over-rule a researcher's self-determination of NHSR.

b. Principal Investigators (PI) may request EDRO documentation, rather than official IRB documentation, that a proposed activity is Not Human Subjects Research (NHSR) provided the activity:

1. falls entirely outside of NASA and U.S. Department of Health and Human Services (HHS) Common Rule regulatory definitions of Research with Human Subjects, and Food and Drug Administration (FDA) definition for clinical investigation; and
2. does not involve International Partner (IP) research, reviewed by another international ethics board; and

i. IP research is assessed for a NHSR determination case-by-case and may deviate from this policy as determined by the NASA IRB Chair

2. An NHSR determination applies only to IRB oversight and does not absolve requirements from other offices or boards (e.g., Safety Office, Crew Office, LSAH/LSDA) and/or requirements from NASA Office of General Counsel. Fulfillment of any additional requirements are the responsibility of the PI.
    a. For projects involving astronaut subjects the PI must coordinate with the NASA Crew Office ***regardless of*** an NHSR determination from the NASA IRB.
    b. Projects involving International Partners (IP) continue to require review by the HRMRB and other international ethics boards as appropriate.
3. PI continue to be responsible for:
    a. Compliance with all applicable federal and state laws.
    b. Ethical collection and use of information used in the project.
    c. Adherence to applicable NASA policies.
4. An NHSR determination applies only to the activities as initially described. A new determination should be made prior to implementation of changes.

Procedures

1. Principal Investigator (PI) initial assessment

a. PI should review the definitions and examples in this document and guidance HRP-415 to determine whether an activity meets the regulatory definitions for human subjects research.

b. PI may complete the 'Not Human Subjects Research (NHSR) Questionnaire' (Appendix 1) to assess their planned activities. PI are encouraged to consult with NASA IRB staff whenever there is uncertainty.

   i) NASA IRB staff may advise PI informally by email or telephone; and/or

   ii) NASA IRB staff may request submission of a 'Not Human Subjects Research (NHSR) Questionnaire' (Appendix 1) through the eIRB to initiate a formal IRB determination process.

   iii) NASA IRB Staff, the NASA IRB Chair, or the convened NASA IRB may determine that the project constitutes Human Subjects Research (HSR) for which IRB oversight is required. The NASA IRB staff may request submission of a protocol and supporting documents through the eIRB for NASA IRB review (Exempt, Expedited, or Full Board).

   iv) PI are encouraged to retain documentation of the relevant factors impacting initial assessments of NHSR (e.g., retention of a completed NHSR Questionnaire within the project records).

2. <u>EDRO review and documentation</u>

   a. Projects meeting criteria for EDRO review (Policy 1.b above) should follow EDRO procedures for requesting EDRO documentation.

3. <u>NASA IRB NHSR determination</u>

   a. PI may initiate an official determination made by the NASA IRB staff through submission of Appendix 1: 'Not Human Subjects (NHSR) Questionnaire'

   b. NASA IRB staff review of the NHSR Questionnaire occurs within the eIRB system.

   c. PI initiates the formal NHSR review in the eIRB via "Create New Study" process.

      i) Basic study information (i.e., title, brief description) is entered in the eIRB smartform.

      ii) The NHSR Questionnaire is attached in place of the protocol document.

      iii) PIs do not need to list study team members or upload COI/CITI documentation for an NHSR determination

      iv) Informed consent and recruitment materials are <u>not</u> required for an NHSR determination. If consent or recruitment materials are submitted, they should be attached to the Local Site Documents section of the eIRB smartfom.

      v) Data collection materials (e.g., surveys, questionnaires) are not typically required as part of an NHSR submission but may be requested by IRB staff when making their determination. These materials may be attached to the Local Site Documents section of the eIRB smartform.

      vi) Documentation from Life Sciences Data Archive (LSDA) or Lifetime Surveillance of Astronaut Health (LSAH) that data can be provided to the PI de-identified/non-attributable must be attached for projects involving secondary data obtained from LSDA or LSAH.

         (1) If LSDA/LSAH cannot provide data in a de-identified/non-attributable format, the project is not eligible for NHSR review.

d. The NASA IRB Chair/Designee reviews the NHSR Questionnaire and determines whether the research meets the definitions for research with human subjects.

   i) The form may be returned to the PI for clarifications or to request additional information.

   ii) PI of projects determined to meet the definitions of HSR (i.e., NOT NHSR) are directed to submit a research protocol and supporting documents (e.g., consent forms, recruitment materials, surveys, questionnaires, etc.) to the eIRB for applicable review by the NASA IRB (Exempt, Expedited, or Full Board).

   iii) Projects confirmed as not meeting the definitions for *Research* with *Human Subjects* (i.e., NHSR) receive formal documentation in the form of a letter emailed through the eIRB system.

e. International regulations related to human subjects research may vary. Projects determined not to meet the U.S. regulatory definitions for research with human subjects may still be considered HSR subject to oversight by other International Partner (IP) agencies.

   i) All projects conducted on ISS must also be reviewed by the Human Research Multilateral Review Board (HRMRB).

   ii) HRMRB review is initiated after an IRB determination letter is issued.

   iii) Responsibility for obtaining approval from other IP agencies.

      (1) NASA HRP funded studies – ROI facilitates approval from other IP agencies.

      (2) Studies that are not funded by HRP – IP approval is the responsibility of the PI

4. Informed Consent

   a. Domestic NHSR projects, conducted by U.S. PI, involving U.S. subjects only:

      i) U.S. regulations governing human subjects do not *require* an informed consent process for NHSR; however, best practice for ethical work with human participants includes informing them of the project's purpose, what they are asked to do, expected duration, associated risks, etc. of their participation.

      ii) PI are responsible for ensuring that any non-IRB related consent/ permission/agreement(s) are obtained and documented as required.

   b. NHSR projects involving international partner (IP) as PI or subjects and/or projects conducted on ISS that intend to enroll US astronauts should add the following language to the Multilateral (HRMRB) Consent Form:

      "**NOTE FOR US CREW ONLY: The NASA IRB has determined the proposed project is not considered human subjects research per U.S. 14CFR1230. As such, 14CFR1230 regulations have not been applied and the project is not under the purview of the NASA IRB."**

Definitions

Applicable to all activities: 2018 Common Rule [14 CFR 1230.102, 45 CFR 46.102]

Generalizable Knowledge – Activities designed to develop or contribute to generalizable knowledge are those designed to draw broadly applicable conclusions or generalize findings

beyond a single individual or internal program, uncover underlying principles that have predictive value and can be applied to other circumstances, or develop or test scientific theories or hypotheses.

Human Subject – a living individual about whom and investigator conducting research [14 CFR1230.102(e)(1)]

i) obtains information or biospecimens through intervention or interaction with the individual and uses, studies, or analyzes the information or biospecimens; or

ii) obtains, uses, studies, analyzes, or generates identifiable private information or biospecimens.

Identifiable private information – private information for which the identity of the subject is or may be readily ascertained (directly or indirectly), by the investigator (or others) or associated with the information. [14 CFR1230.102(e)(5)]

Interaction – communication or interpersonal contact between investigator and subject. [14 CFR1230.102(e)(3)]

Intervention – includes both physical procedures by which information or biospecimens are gathered and manipulations of the subject or the subject's environment that are performed for research purposes. [14 CFR1230.102(e)(2)]

Private information – information about behavior that occurs in a context in which an individual can reasonably expect that no observation or recording is taking place, as well as information that has been provided for specific purposes by an individual and that the individual can reasonably expect will not be made public. [14 CFR1230.102(e)(4)]

Research – a systematic investigation, including research development, testing, and evaluation, designed to develop or contribute to generalizable knowledge. [14 CFR1230.102(l)]

Systematic investigation – Examination that involves a prospective plan that incorporates data collection, either quantitative or qualitative, and data analysis to answer a question.

In addition to the above definitions, activities regulated by the Food and Drug Administration (FDA) also apply the FDA definitions [21 CFR 50.3]

Clinical Investigation -Any experiment that involves a test article and one or more human subjects that is either subject to requirements for prior submission to the Food and Drug Administration under section 505(i), or 520(g) of the act, or is not subject to requirements for prior submission to the Food and Drug Administration under these sections of the act, but the results of which are intended to be submitted to, or held for inspection by the Food and Drug Administration as part of an application for a research or marketing permit." The term does not include experiments that are subject to the provision of 21 CFR 58, regarding nonclinical laboratory studies. [21 CFR 50.3(c)]

Note: Under FDA regulations, the terms "research" and "clinical investigation" are synonymous.

Test Article - A test article means any drug (including a biological product for human use), medical device for human use, human food additive, color additive, electronic product, or any other article subject to regulation under the Federal Food, Drug and Cosmetic Act [21 CFR 50.3(j)].

Human Subject (FDA) - an individual who is or becomes a participant in research, either as a recipient of the test article or as a control. A subject may be either a healthy human or a patient. [21 CFR 50.3(g)]

Acronyms

FDA – US Food and Drug Administration

HHS – US Department of Health and Human Services

HITL – Human in the loop

HRMRB – Human Research Multilateral Review Board

HSR – Human Subjects Research

IRB – Institutional Review Board

PI – Principal Investigator

NASA – National Aeronautics and Space Administration

NHSR – Not Human Subjects Research

ROI – Research Operations and Integration

References

NASA

NASA 14 CFR 1230.102
NPD 7100
NASA NPD 7100

HHS Regulations

HHS 45CFR 46.102
Engagement of Institutions in Human Subjects Research (2008)
OHRP 'Human Subject Regulations Decision Charts: 2018 Requirements, Chart 1
OHRP Draft Guidance: Activities Deemed Not to Be Research: Public Health Surveillance 2018 Requirements

FDA Regulations

21CFR50.3(c)(g)
21CFR56.103(c)(e)
21CFR312.3(b)
21CFR812.3(h)(p)

Version #	Version Notes:	Originated By:	Approved By:	Date Approved:
v.7.1	Minor revisions (i.e., removal of HHS references, reformat)	J. Ensley Gorshe	M. Covington	14Aug2023
v.7	Major revisions to regulatory interpretations	J. Ensley Gorshe	M. Covington	19May2022

v.2-v.6		J. Kisenwether	M. Covington	27Sept2021
v.1	Original	J. Christensen/ J. Kisenwether	M. Covington	21Oct2020

NASA HRP-405 Appendix 1

Not Human Subjects Research (NHSR) Questionnaire

Date: Click or tap to enter a date.

General Information

Research Title:	Click or tap here to enter text.
PI:	Click or tap here to enter text.
NASA Center (if applicable):	Click or tap here to enter text.
Research location:	Click or tap here to enter text.
Funding source:	Click or tap here to enter text.

This form is for use when determining whether a project meets the regulatory definitions for Human Subject Research

Prospective review and approval by an IRB is required for all activities that meet the regulatory definitions for *research* with *human subjects* (Common Rule) or a *clinical investigation* (FDA). Activities that do not meet the regulatory definitions are considered "Not Human Subjects Research" (NHSR) for which IRB oversight is not required.

Section I: Regulatory Definitions of *Research*

Section I: Regulatory Definitions of Research

Research is defined as "*a systematic investigation, including research development, testing and evaluation, designed to develop or contribute to generalizable knowledge" [NASA 14CFR1230. 102(l); HHS 45CFR46.102(l)]*

Under FDA regulations, the terms "Research" and "Clinical Investigation" and "Clinical Study" are synonymous and are defined as:

> *"Any experiment that involves a test article* and one or more human subjects that is either subject to requirements for prior submission to the Food and Drug Administration under section 505(i), or 520(g) of the act, or … results of which are intended to be submitted to, or held for inspection by the Food and Drug Administration as part of an application for a research or marketing permit." FDA 21CFR50.3(c)*

**A test article means any drug (including a biological product for human use), medical device for human use, human food additive, color additive, electronic product, or any other*

article subject to regulation under the Federal Food, Drug and Cosmetic Act (21 CFR 50.3(j)).

1.1 What are the aims of the proposed project or activities?

Click or tap here to enter text.

1.2 Systematic Investigation.

Does the project or activity involve a systematic investigation, including research development, testing, or evaluation?

A systematic investigation is an activity that involves a prospective plan to gather information (i.e., data collection (quantitative or qualitative)) and analysis to answer a question.

☐ **Yes.** The project/activity involves a *systematic investigation.*
☐ **No.**
☐ **Unsure.**

1.3 Designed to contribute to Generalizable Knowledge.

Is the project/activity designed to develop or contribute to generalizable knowledge?

Activities designed to develop or contribute to generalizable knowledge are those designed to draw broadly applicable conclusions or generalize findings beyond a single individual or internal program, uncover underlying principles that have predictive value and can be applied to other circumstances, or develop or test scientific theories or hypotheses.

> *NOTE: Although publication is often viewed as evidence of research status, it is not the only criterion. In fact, "systematic investigations" often result in published information, yet they do not qualify as research because they were not designed to contribute to generalizable knowledge.*

☐ **Yes.** The proposed project/activity is *designed to contribute to generalizable knowledge.*
☐ **No.**
☐ **Unsure.**

1.4 FDA Clinical Investigation.

Does the project/activity involve a test article subject to FDA regulation (e.g., drug, biologic, medical device, food additive, etc.) **which requires prior submission to the FDA, or for which the results will be submitted to the FDA as part of a research or marketing permit?**

☐ **Yes.** The proposed activities are subject to FDA regulation.
☐ **No.**
☐ **Unsure.**

Checklist: Definition of Research Use responses to questions 1.1-1.4 to complete the checklist.	Yes	No	Not Sure
Is the project a ***systematic investigation*** (1.2) including research development, testing, and evaluation, designed to develop or contribute to ***generalizable knowledge***(1.3)?	☐	☐	☐
Does the project involve a research/clinical investigation regulated by the FDA (1.4)?	☐	☐	☐

If you answered "NO" to both of the above, STOP the project is NOT Human Subjects Research and IRB oversight is not necessary

- Please retain this worksheet as documentation with project files.
- If a formal determination from the NASA IRB is needed please submit this form to the NASA eIRB to receive formal documentation that the project is "Not Human Subjects Research"

If you answered "YES" or "NOT SURE" to either of the above in Section I: Definition of Research à proceed to Section II: Definition of Human Subject

Section II: Definition of Human Subject

Section II: Definition of Human Subject Research

Human Subject, defined as "*a living individual about whom an investigator (whether professional or student) conducting research:*

(1) Obtains information or biospecimens through intervention or interaction with the individual, and uses, studies, or analyzes the information or biospecimens; ***OR***

(2) Obtains, uses, studies, analyzes, or generates identifiable private information or biospecimens." *[NASA 14CFR1230.102(e); HHS 45CFR46.102(e)]*

FDA Human Subject defined as *"an individual who is or becomes a participant in research, either as a recipient of the test article or as a control. A subject may be either a healthy human or a patient."* [FDA 21CFR50.3(g), 21CFR56.103(e), 21CFR312.3(b), and 21CFR812.3(p)].

2.1 Describe the participants or population you intend to include the project.

Click or tap here to enter text.

2.1.a Does the population involve International Partners or astronaut crew?

☐ **No.** No astronaut or International Partner participants.
☐ **Yes.**

If yes, please describe:

Click or tap here to enter text.

2.2 Do the information or biospecimens collected or used for the project/activity involve living individual(s)?

☐ **No.** Individuals are NOT living – STOP. The project is NHSR.
☐ **Yes.** Individuals ARE living.

2.3 Does the project/activity:

(i) Obtain information or biospecimens through intervention or interaction with the individual(s), and use, study, or analyze the information or biospecimens; [HHS and NASA]

(ii) Obtain, use, study, analyze, or generate identifiable private information or identifiable biospecimens? [HHS and NASA]

or

(iii) Use the individual as a recipient of a test article or as a control? [FDA]

Intervention *– includes both physical procedures by which information or biospecimens are gathered (e.g., venipuncture) and manipulations of the subject or subject's environment that are performed for research purposes.*

Interaction *– includes communication or interpersonal contact between investigator and subject.*

Private information *– includes information about behavior that occurs in a context in which an individual can reasonably expect that no observation or recording is taking place and information that has been provided for specific purposes by an individual and that individual can reasonably expect will not be made public (e.g., medical record).*

Identifiable private information/biospecimen *– information/biospecimen for which the identify of the subject is or may readily be ascertained by someone on the research team or associated with the information/biospecimen.*

Test article *– means any drug, medical device, human food additive, color additive, electronic product or other article subject to regulation by the FDA.*

2.3.a What are the procedures?

Click or tap here to enter text.

2.3.b If the project/activity only involves secondary use of data and/or biospecimens (i.e., no interaction or intervention with participations):

(1) **Describe the origin of the data/specimens** (i.e., purchased from a biobank, MedB data, existing data from another project):

Click or tap here to enter text.

(2) **Address whether the data/specimens will identifiable**[1]

Click or tap here to enter text.

Checklist: Definition of Human Subjects Use responses to questions 2.1-2.3 to complete the checklist.		Yes	No	Not Sure
	Are the information or biospecimens obtained or used for the research **about living individuals?**	☐	☐	☐
	If you answered "YES" or "NOT SURE" please continue with the questions. If you answered "NO", the activity is NOT Human Subject Research, no IRB review is needed			
	Does the research involve **interaction** or **intervention** with humans?	☐	☐	☐

[1] Identifiable: The identity of the subject is or may readily be ascertained by the investigator or associated with the information. This includes private information or specimens linked to specific individuals by the investigator(s) either directly or indirectly through coding systems. Includes when individuals can be indirectly identified through a combination of information included in the data.

Does the research involve Human Subjects?	Does the research obtain, use, study, analyze, or generate identifiable **private information** or **identifiable biospecimens**?	☐	☐	☐
	Does the research involve human subjects (healthy or patients) who will serve as the recipient or a **test article** or as a control?	☐	☐	☐

If you answered "YES" to any of the three questions above **the activity is Human Subject Research**, **an eIRB submission is required.**

If you answered "NO" to all three questions above, the activity is NOT Human Subject Research **Submit this form through the eIRB to receive official IRB documentation.**

If you answered "NOT SURE" to any of the above, **please submit this form to the NASA eIRB and contact the NASA IRB Office.**

3. Additional information, if any, that may be helpful to the NASA IRB in making this determination.

Click or tap here to enter text.